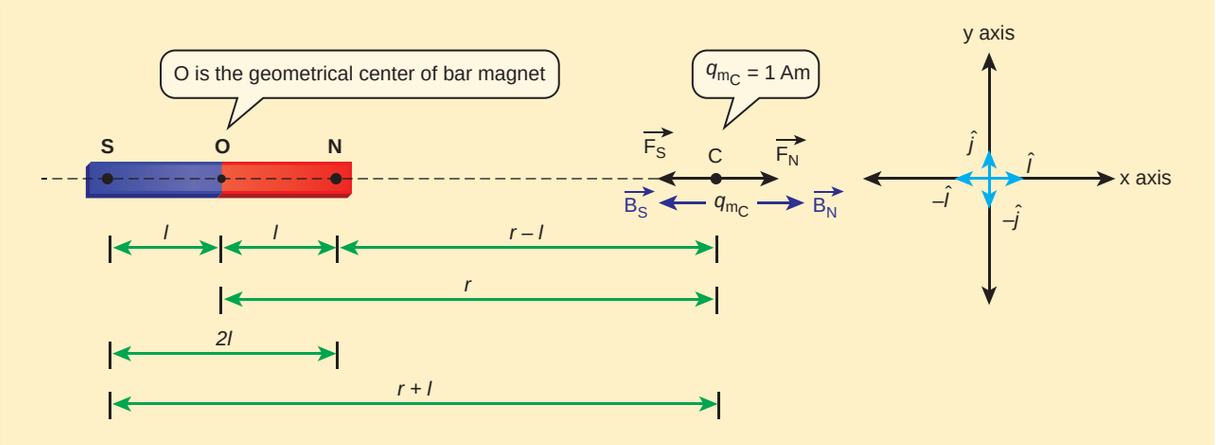


Figure 3.16 Magnetic field at a point along the axial line due to magnetic dipole

From equation (3.9) and (3.10), the net force at point C is $\vec{F} = \vec{F}_N + \vec{F}_S$. From definition, this net force is the magnetic field due to magnetic dipole at a point C $\left(\vec{F} = \vec{B}\right)$

$$\vec{B} = \frac{\mu_\circ}{4\pi}\frac{q_m}{(r-l)^2}\hat{i} + \left(-\frac{\mu_\circ}{4\pi}\frac{q_m}{(r+l)^2}\hat{i}\right)$$

$$\vec{B} = \frac{\mu_\circ q_m}{4\pi}\left(\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2}\right)\hat{i}$$

$$\vec{B} = \frac{\mu_\circ 2r}{4\pi}\left(\frac{q_m \cdot (2l)}{\left(r^2 - l^2\right)^2}\right)\hat{i} \qquad (3.11)$$

Since, magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m \cdot 2l$ the magnetic field at a point C equation (3.11) can be written as

$$\vec{B}_{axial} = \frac{\mu_\circ}{4\pi}\left(\frac{2rp_m}{\left(r^2 - l^2\right)^2}\right)\hat{i} \qquad (3.12)$$

If the distance between two poles in a bar magnet are small (looks like short magnet) compared to the distance between geometrical centre O of bar magnet and the location of point C i.e., r >>l then,

$$(r^2 - l^2)^2 \approx r^4 \qquad (3.13)$$

Therefore, using equation (3.13) in equation (3.12), we get

$$\vec{B}_{axial} = \frac{\mu_\circ}{4\pi}\left(\frac{2p_m}{r^3}\right)\hat{i} = \frac{\mu_\circ}{4\pi}\frac{2}{r^3}\vec{p}_m \qquad (3.14)$$

where $\vec{p}_m = p_m\hat{i}$.

3.2.2. Magnetic field at a point along the equatorial line due to a magnetic dipole (bar magnet)

Consider a bar magnet NS as shown in Figure 3.17. Let N be the north pole and S be the south pole of the bar magnet, each with pole strength q_m and separated by a distance of 2l. The magnetic field at a point C (lies along the equatorial line) at a distance r from the geometrical center O of the bar magnet can be computed by keeping unit north pole (q_{mC} = 1 A m) at C. The force experienced by the unit north pole at C due to pole strength N-S can be computed using Coulomb's law of magnetism as follows:

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